



A4 Diversity of organisms

A · Unity and diversity · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Organisms are named using binomial nomenclature (genus + species) and classified hierarchically, with molecular evidence (especially rRNA gene sequences) placing all life into three domains: Archaea, Bacteria and Eukarya. A species is defined as a group of organisms able to interbreed and produce fertile offspring in natural conditions. Phylogenetic trees (cladograms) are built from shared derived characteristics, including DNA/protein sequence similarity, to represent evolutionary relationships, with a clade containing a common ancestor and all of its descendants.

Worked example: follow the reasoning

Four organisms are compared for four characteristics; organisms sharing the most derived characteristics are grouped as most closely related. Suppose organisms P and Q share 3 of 4 derived characteristics, while P and R share only 1. Step 1: count shared derived (not ancestral) characters between each pair. Step 2: the pair sharing the most derived characters (P and Q, sharing 3) is placed as most closely related, branching from a more recent common ancestor than P and R. Conclusion: P and Q should be drawn as sister taxa on the cladogram, with R branching off earlier.

Quick check · multiple choice

What is the main basis for classifying all living organisms into the three-domain system (Archaea, Bacteria, Eukarya)?

- A Similarity in outward morphology only
- B Differences in rRNA gene sequence and cell biochemistry
- C Ability of organisms to interbreed
- D Similarity of habitat

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A biologist compares four bacterial-like organisms and finds the following counts of shared derived molecular characteristics with a reference sequence: Organism W: 92% similarity; Organism X: 76% similarity; Organism Y: 98% similarity; Organism Z: 60% similarity. [7]
 - (a) State which organism is most closely related to the reference organism, and justify your answer. [2]
 - (b) Explain why rRNA sequences, rather than a single visible physical feature, are considered reliable evidence for constructing phylogenies across all three domains of life. [3]
 - (c) Sketch (describe in words) how W, X, Y and Z would be arranged on a cladogram relative to the reference organism, from most to least closely related. [2]

Final answer: _____



Answers and explanation

Question 1 [7]

(a) Organism Y, because it has the highest percentage sequence similarity (98%) to the reference organism, indicating the fewest accumulated differences and therefore the most recent common ancestor.

(b) rRNA genes are present in all known organisms (bacteria, archaea and eukaryotes), are highly conserved due to their essential function, and accumulate mutations at a relatively steady rate over time, so the degree of sequence difference can be used as a molecular clock to estimate evolutionary relatedness independent of superficial physical similarity, which can arise from convergent evolution rather than shared ancestry.

(c) From most to least closely related to the reference: Y (98%), W (92%), X (76%), Z (60%); on the cladogram, Y branches off closest to the reference organism (most recent shared node), followed by W, then X, then Z branching off earliest (most distant common ancestor).

Quick check answer

B. Comparing highly conserved rRNA gene sequences (and associated biochemical/cellular features) across organisms revealed three fundamentally distinct evolutionary lineages, leading to the three-domain system.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Biology HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.